Hierarchy-only engineering candidate. Signed Review-A pin/net authority is retained; routing, DRC, DFM, Review B and manufacture remain prohibited.

Power entry and rail interface

3V3_MIC
3V3_MODEM
EN_MODEM
GND_DIGITAL
GND_MODEM
I2C2_SDA_BUS
PWR_GOOD
3V3_DIGITAL
FAULT
GND_MIC
I2C2_SCL_BUS
PWR_FAULT

Файл: PCB-MAIN_01_POWER.kicad_sch

GNSS timing antenna and supervisor

3V3_DIGITAL
GNSS_PPS_U1
GNSS_TX_U1
GND_DIGITAL
GNSS_SPI_U5

Файл: PCB-MAIN_04_GNSS.kicad_sch

BLE module reset DFU and SWD

3V3_DIGITAL
BLE_EN
BLE_TX_U1
BLE_RFU_REQ
BLE_SPI_U5
GND_DIGITAL

Файл: PCB-MAIN_07_BLE.kicad_sch

MCU clocks reset and straps

3V3_DIGITAL
ACCEL_INT
BLE_EN
BLE_TX_U1
CELL_DTR_U16
CELL_RESET_N_CMD
CELL_RX_U16
CELL_TX_U16
EN_MODEM
GNSS_PPS_U1
GND_TX_U1
I2C2_SDA_U1
LORA_BUSY
LORA_DIO1
LORA_MOSI_U1
LORA_RESET_N
LORA_SCK_U1
MCU_WAKE
NOR_CDS_U1
NOR_CD2_U1
NOR_DCS_U2
PDM_CLK
PDM_DATA2
PDM_DATA4
PWR_GOOD
REV_STRAP1
SD_CMD_U1
SD_D1_U1
SD_D3_U1
SIM_DET
SIM_RUIM_EN
SWCLK
TAMPER_IN_U1
TEST_UART_TX_U1
USB_DP_U1

Файл: PCB-MAIN_02_MCU.kicad_sch

Cellular modem dual SIM and recovery

AUX_CFG
BLE_RFU_REQ
BLE_SPI_U5
BUSY0
CELL_POWER_CMD
CELL_SPI_U16
CELL_STATUS_U16
EN_MODEM
GND_DIGITAL
GNSS_SPI_U5
I2C2_SCL_U5
LORA_BUSY
LORA_DIO5_U5
LORA_MOSI_U5
LORA_RXEN
LORA_TXEN
NOR_SPI_U5
NOR_CDS_U5
NOR_CD3_U5
RST
PDM_DATA5
PWR_FAULT
REV_STRAP0
SD_DM_U5
SD_DP_U5
SD_SD_U5
SIM_DET
SIM_RUIM_SEL
SWDIO
TEST_UART_RX_U5
USB_DM_U5
USB_VBUS_SENSE

Файл: PCB-MAIN_05_CELLULAR.kicad_sch

Storage sensors microSD and tamper

3V3_MODEM
3V3_DIGITAL
CELL_POWER_CMD
GND_DIGITAL
I2C2_SCL_U1
I2C2_SDA_U1
CELL_SPI_U16
NOR_CDS_U1
NOR_CD2_U1
CELL_STATUS_U16
NOR_CDS_U2
GND_MODEM
SD_CMD_U1
SIM_DET
SD_D1_U1
SIM_RUIM_SEL
SD_D3_U1
TAMPER_IN_U1
ACCEL_INT
I2C2_SCL_BUS
I2C2_SDA_BUS
NOR_SPI_U5
NOR_CDS_U5
SD_DM_U5
SD_DP_U5
SD_SD_U5
SIM_DET

Файл: PCB-MAIN_08_STORAGE_SENSORS.kicad_sch

PDM audio AAD and microphone harnesses

3V3_MIC
AUX_CFG
GND_MIC
MCU_WAKE
PDM_CLK
PDM_DATA2
PDM_DATA4
3V3_DIGITAL
GND_DIGITAL
PDM_DATA5

Файл: PCB-MAIN_03_AUDIO.kicad_sch

LoRa radio control and conducted RF

3V3_DIGITAL
LORA_BUSY
GND_DIGITAL
LORA_DIO5_U1
LORA_MOSI_U1
LORA_SCK_U1
LORA_RESET_N
LORA_RXEN
LORA_TXEN
GND_DIGITAL
LORA_MOSI_U5
LORA_SCK_U5

Файл: PCB-MAIN_06_LORA.kicad_sch

USB service debug and EOL fixture

3V3_MIC
3V3_MODEM
RST
I2C2_SCL_BUS
RST
REV_STRAP0
SWCLK
TEST_UART_TX_U1
USB_DM_U1
USB_VBUS_SENSE
3V3_DIGITAL
BUSY0
GND_DIGITAL
I2C2_SDA_BUS
PWR_GOOD
REV_STRAP1
SWDIO
TEST_UART_RX_U5
USB_DP_U5

Файл: PCB-MAIN_09_CONNECTORS_TEST.kicad_sch

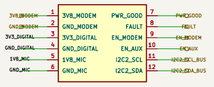
NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Root sheet 1 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced
Dioneya / ZS-BPLA
Sheet: /
File: PCB-MAIN.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — System overview

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

J_PWR



C19

43045-1202



22 uf 10 V XSR

C20



10 uf 10 V XSR

R103



0 Ohm

NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 2 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /Power entry and rail interface/
File: PCB-MAIN_01_POWER.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — Power

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

U1

STM32U585VIT6Q

X1

32.768 kHz full-swing LVCMOS

C1

100 nF 25 V X7R

C2

100 nF 25 V X7R

C3

100 nF 25 V X7R

C4

100 nF 25 V X7R

C5

100 nF 25 V X7R

C6

10 uF 10 V X5R

C7

100 nF 25 V X7R

C8

100 nF 25 V X7R

C9

1 uF 16 V X5R

C10

100 nF 25 V X7R

C11

1 uF 16 V X5R

C12

100 nF 25 V X7R

C13

10 uF 10 V X5R

C14

2.2 uF 10 V X5R

C15

2.2 uF 10 V X5R

C16

100 nF 25 V X7R

C17

100 nF 25 V X7R

C18

100 nF 25 V X7R

L1

2.2 uH 20%

R1

10 kOhm 1%

R2

100 kOhm 1%

R3

100 kOhm 1%

R4

100 kOhm 1%

R5

100 kOhm 1%

R6

100 kOhm 1%

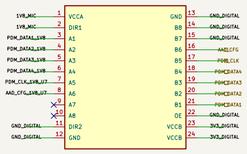
NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 3 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced
Dioneya / ZS-BPLA
Sheet: /MCU clocks reset and straps/
File: PCB-MAIN_02_MCU.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — MCU

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

U7



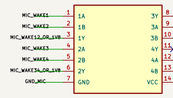
SN74AXC8T245PWR

J_MIC4



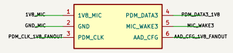
504050-0691

U17



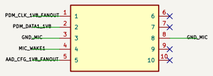
SN74LVC32APWR

J_MIC3



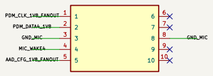
504050-0691

U19



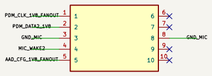
4 channel 0.5 pF ESD

U22



4 channel 0.5 pF ESD

U20



4 channel 0.5 pF ESD

J_MIC1



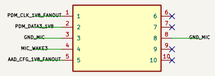
504050-0691

U18



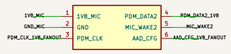
C32 SN74AXC1T45DRLR

U21



4 channel 0.5 pF ESD

J_MIC2



504050-0691

C27



100 nF 25 V X7R

C28



100 nF 25 V X7R

C29



100 nF 25 V X7R

C30



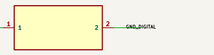
100 nF 25 V X7R

C31



100 nF 25 V X7R

C32



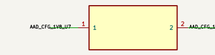
100 nF 25 V X7R

R17



22 Ohm 1%

R18



22 Ohm 1%

R19



100 kOhm 1%

R20



100 kOhm 1%

R21



100 kOhm 1%

R22



100 kOhm 1%

R23



100 kOhm 1%

NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 4 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /PDM audio AAD and microphone harnesses/
File: PCB-MAIN_03_AUDIO.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — PDM audio

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

U9



U5



Single rail-to-rail low-power op amp

Q4



Complementary N and P channel 20 V

FL1



1582.4 MHz wideband GNSS SAW

J9



U.FL-R-SMT-1(60)

C60



100 nF 25 V X7R

R56



22 Ohm 1%

MAX-M10S-00B

C61



10 uF 10 V X5R

R57



22 Ohm 1%

C62



100 nF 25 V X7R

R58



22 Ohm 1%

C63



10 nF 16 V X7R

R59



560 Ohm 5% 0.1 W

C64



47 pF 25 V C0G

R60



100 kOhm 5% 0.1 W

C65



100 nF 25 V X7R

R61



100 kOhm 5% 0.1 W

D4



5.5 V unidirectional 0.5 pF RF ESD

R62



10 Ohm 5% 0.25 W

L2



27 nH 5%

NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 5 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced

Dioneya / ZS-BPLA

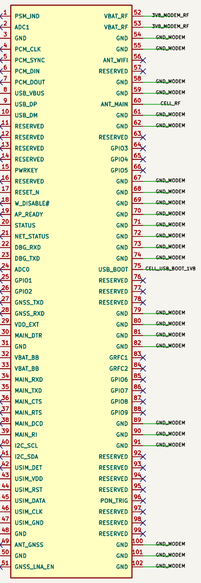
Sheet: /GNSS timing antenna and supervisor/
File: PCB-MAIN_04_GNSS.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — GNSS

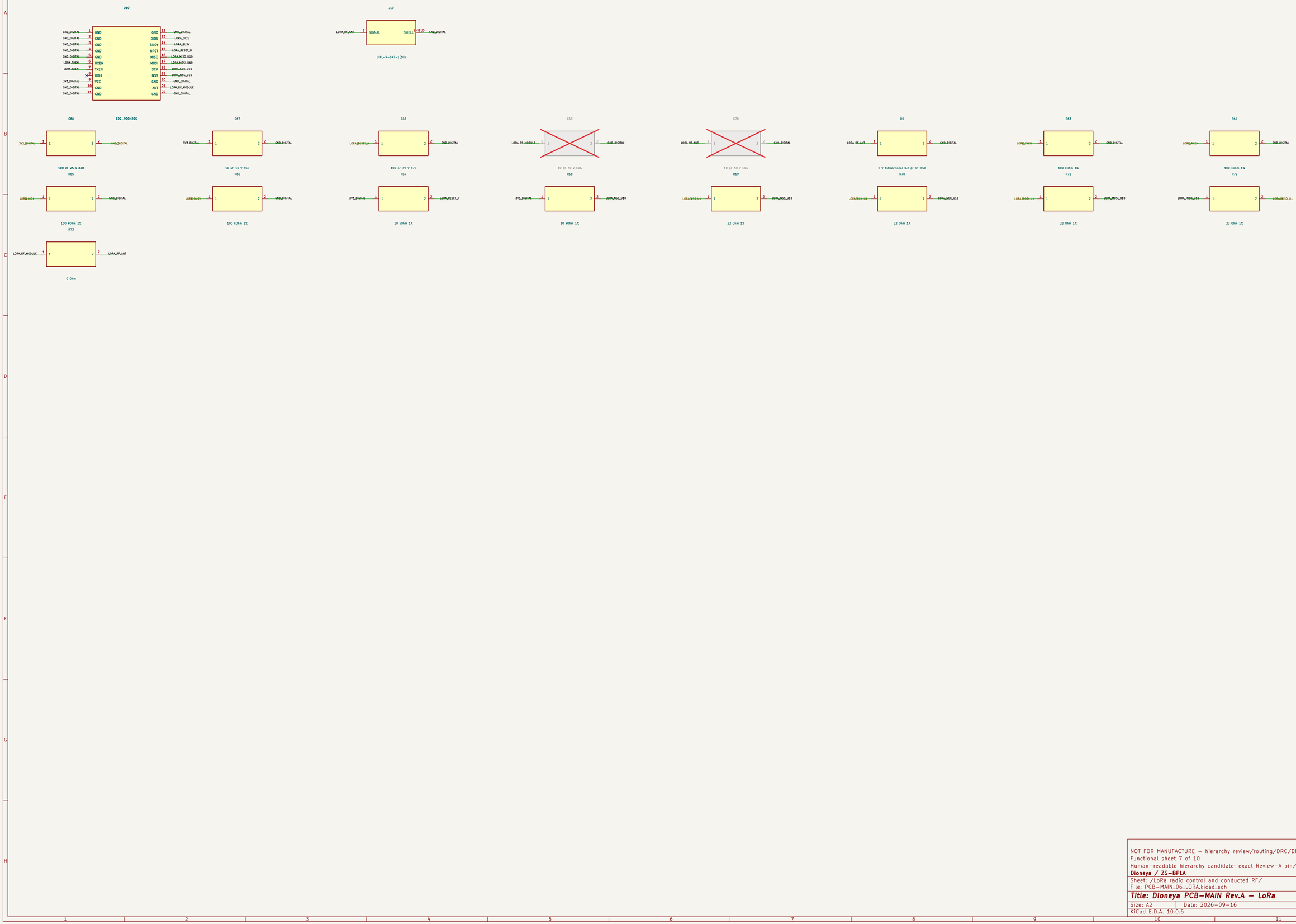
Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

U8


PSM_IND
ADC1
GND
PCM_CLK
PCM_SYNC
PCM_DIN
PCM_DOUT
USB_VBUS
USB_DP
USB_DM
RESERVED
RESERVED
RESERVED
RESERVED
PWRKEY
RESERVED
RESET_N
W_DISABLE#
AP_READY
STATUS
NET_STATUS
DBG_RXD
DBG_TXD
ADC0
GPIO1
GPIO2
GNSS_TXD
GNSS_RXD
VDD_EXT
MAIN_DTR
GND
VBAT_BB
VBAT_BB
MAIN_RXD
MAIN_TXD
MAIN_CTS
MAIN_RTS
MAIN_DCD
MAIN_RI
I2C_SCL
I2C_SDA
RESERVED
USIM_DET
USIM_VDD
USIM_RST
USIM_DATA
USIM_CLK
USIM_GND
GND
ANT_GNSS
GND
GNSS_LNA_EN
VBAT_RF
VBAT_RF
GND
GND
ANT_MAIN
GND
GND
RESERVED
GPIO3
GPIO4
GPIO5
GND
GND
GND

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.



NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 7 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced

Dioneya / ZS-BPLA

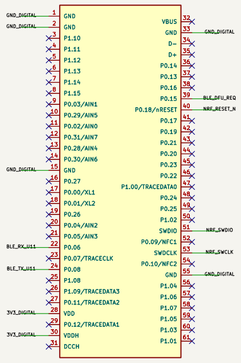
Sheet: /LoRa radio control and conducted RF/
File: PCB-MAIN_06_LORA.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — LoRa

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

U11



U6



Single non-inverting open-drain buffer

TP_BLE_SWD



TEST_PADS

C71 MDBT50Q-P1MV2



100 nF 25 V X7R

C72



10 uF 10 V X5R

C73



100 nF 25 V X7R

R74



22 Ohm 1%

R75



22 Ohm 1%

R76



10 kOhm 1%

R77



100 kOhm 1%

R78



10 kOhm 1%

NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 8 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced
Dioneya / ZS-BPLA
Sheet: /BLE module reset DFU and SWD/
File: PCB-MAIN_07_BLE.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — BLE

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

U2

W25Q512JVFIQ

U3

LIS2DW12TR

J12

MEM2052-00-195-00-A

U23

4 channel 0.5 pF ESD

U24

4 channel 0.5 pF ESD

U12

SDCIT2/32GB

U4

STTS22HTR

J13

504050-0291

C21

100 nF 25 V X7R

C76

10 nF 25 V X7R

R11

22 Ohm 1%

RB1

22 Ohm 1%

RB9

10 kOhm 1%

C22

1 uF 16 V X5R

C78

10 nF 25 V X7R

R12

22 Ohm 1%

RB2

22 Ohm 1%

RB10

10 kOhm 1%

C23

100 nF 25 V X7R

D9

5.5 V 0.5 pF ESD

R13

2.2 kOhm 1%

RB3

22 Ohm 1%

R99

10 kOhm 1%

C24

10 uF 10 V X5R

D11

5.5 V 0.5 pF ESD

R14

2.2 kOhm 1%

RB4

22 Ohm 1%

R100

1 kOhm 1%

C25

100 nF 25 V X7R

R7

10 kOhm 1%

R15

22 Ohm 1%

RB5

10 kOhm 1%

C26

100 nF 25 V X7R

R8

22 Ohm 1%

R16

22 Ohm 1%

RB6

10 kOhm 1%

C74

100 nF 25 V X7R

R9

22 Ohm 1%

R79

22 Ohm 1%

RB7

10 kOhm 1%

C75

10 uF 10 V X5R

R10

22 Ohm 1%

RB0

22 Ohm 1%

RB8

10 kOhm 1%

NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 9 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced

Dioneya / ZS-BPLA

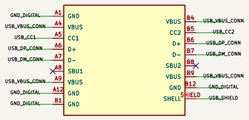
Sheet: /Storage sensors microSD and tamper/
File: PCB-MAIN_08_STORAGE_SENSORS.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — Storage + sensors

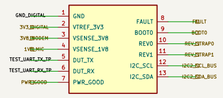
Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/10

Every connected symbol pin has one explicit wire stub. Only hierarchical labels cross functional-sheet boundaries.

J11



TP_EOL



TEST_PADS

TP_MCU_SWD



TEST_PADS

U25



2 channel 0.7 pF USB ESD

C77

USB4105-GF-A-120



4.7 nF 100 V X7R

R95



1 kOhm 1%

D6



5.5 V 0.5 pF ESD

R96



100 kOhm 1%

D7



5.5 V 0.5 pF ESD

R97



1 MOhm 1%

D8



5.5 V 0.5 pF ESD

R98



0 Ohm

R91



22 Ohm 1%

R101



22 Ohm 1%

R92



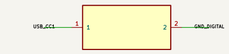
22 Ohm 1%

R102



1 kOhm 1%

R93



5.1 kOhm 1%

R94



5.1 kOhm 1%

NOT FOR MANUFACTURE — hierarchy review/routing/DRC/DFM/Review B pending
Functional sheet 10 of 10
Human-readable hierarchy candidate; exact Review-A pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /USB service debug and EOL fixture/
File: PCB-MAIN_09_CONNECTORS_TEST.kicad_sch

Title: Dioneya PCB-MAIN Rev.A — USB + debug

Size: A2	Date: 2026-09-16	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 10/10